

REGULARISATION IN GUIDING

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1. SETUP

We consider the Markov kernel $P(x_0, dx_1)$ defined by

$$X_1 \mid X_0 \sim \mathcal{N}(\omega + \psi X_0, \eta^2).$$

We assume $v \sim x + \log Z^2$, where $Z \sim \mathcal{N}(0, 1)$. Define

$$h(x_1) = \text{density of } X_1 + \log Z^2 \text{ evaluated at } v, \quad \text{where } Z \sim \mathcal{N}(0, 1).$$

We have

$$P(X_1 + \log Z^2 \leq v) = P(Z^2 \leq e^{v-X_1})$$

so

$$h(x_1) = f_{\chi^2(1)}(e^{v-x_1}) e^{v-x_1}.$$

For large x_1 it will behave like $\sqrt{e^{v-x_1}}$. It should also behave this way for large x_1 . Define, using Gaussian approximation to the distribution of $\log Z^2$

$$g(x_1) = \varphi(v; \mu + x_1, \sigma^2), \quad \mu = -1.27, \quad \sigma^2 = \pi^2/2,$$

where $\varphi(\cdot; \mu, \sigma^2)$ denotes the Gaussian density.

We have twisted kernels

$$P^h(x_0, dx_1) = \frac{P(x_0, dx_1) h(x_1)}{\int P(x_0, dx_1) h(x_1)}, \quad P^g(x_0, dx_1) = \frac{P(x_0, dx_1) g(x_1)}{\int P(x_0, dx_1) g(x_1)}.$$

Define $c_g = \int P(x_0, dx_1) g(x_1)$ and similarly c_h . As x_0 is fixed in this section, we omit it from our notation. In our specific case

$$c_g = \varphi(v; \mu + \omega + \psi x_0; \sigma^2 + \eta^2).$$

Note

$$P^{g+\varepsilon}(x_0, dx_1) = \frac{g(x_1) + \varepsilon}{c_g + \varepsilon} P(x_0, dx_1) = \lambda_g(\varepsilon) P^g(x_0, dx_1) + (1 - \lambda_g(\varepsilon)) P(x_0, dx_1),$$

with $\lambda_g(\varepsilon) = c_g / (c_g + \varepsilon)$.

2. SEQUENTIAL MONTE CARLO APPROACH

Suppose we do sequential Monte Carlo to sample from $X_1 \mid v$ and to find the marginal likelihood $p(v)$, which is actually c_h . We use mutation kernel $M = P^{g+\varepsilon}$. Then it follows from the results in Chapter 5 of Chopin-Papaspiliopoulos that the potential should be

$$\begin{aligned} G_\varepsilon(x_1) &:= h(x_1) \frac{P(x_0, dx_1)}{P^{g+\varepsilon}(x_0, dx_1)} = h(x_1) \frac{\int P(x_0, dx_1) g(x_1) dx_1 + \varepsilon}{g(x_1) + \varepsilon} \\ &= h(x_1) \frac{c_g + \varepsilon}{g(x_1) + \varepsilon}. \end{aligned}$$

The estimator for the marginal likelihood is

$$\hat{L} := \frac{1}{N} \sum_{n=1}^N G_\varepsilon(X_1^{(n)}), \quad (X_1^{(n)})_{n=1}^N \sim P^{g+\varepsilon}(x_0, dx_1).$$

For *any* $\varepsilon > 0$

$$\mathbb{E}_{P^{g+\varepsilon}} \hat{L} = \mathbb{E}_{P^{g+\varepsilon}} G_\varepsilon(X_1) = \int h(x_1) \frac{c_g + \varepsilon}{g(x_1) + \varepsilon} \frac{g(x_1) + \varepsilon}{c_g + \varepsilon} P(x_0, dx_1) = c_h.$$

2.1. Analysis of variance of $\hat{L}$. Define

$$m(\varepsilon) = \mathbb{E}_{P^{g+\varepsilon}} [G_\varepsilon(x_0, X_1)]^2.$$

By expanding we get

$$\begin{aligned} m(\varepsilon) &= \int h(x_1)^2 \left(\frac{c_g + \varepsilon}{g(x_1) + \varepsilon} \right)^2 \frac{P(x_0, dx_1) (g(x_1) + \varepsilon)}{c_g + \varepsilon} \\ &= (c_g + \varepsilon) \int \frac{h(x_1)^2}{g(x_1) + \varepsilon} P(x_0, dx_1). \end{aligned}$$

Define

$$I(\varepsilon) = \int \frac{h(x_1)^2}{g(x_1) + \varepsilon} P(x_0, dx_1)$$

and

$$\bar{m}(\varepsilon) = \log m(\varepsilon) = \log(c_g + \varepsilon) + \log I(\varepsilon).$$

Note that we can rewrite to

$$I(\varepsilon) = \int h(x_1) \frac{\int h(x_1) P(x_0, dx_1)}{g(x_1) + \varepsilon} P^h(x_0, dx_1) = c_h \int \frac{h(x_1)}{g(x_1) + \varepsilon} P^h(x_0, dx_1).$$

though we won't use this expression in the numerics.

2.2. Infinite variance case. Suppose $\varepsilon = 0$. Since $c > 0$, finiteness of $\bar{m}(0)$ is equivalent to finiteness of

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1).$$

Using $f_{\chi^2(1)}(u) = (2\pi u)^{-1/2} e^{-u/2}$ we get

$$h(x_1) = \frac{1}{\sqrt{2\pi}} e^{-(v-x_1)/2} e^{-e^{v-x_1}/2},$$

so

$$h(x_1)^2 = \frac{1}{2\pi} e^{-(v-x_1)} e^{-e^{v-x_1}}.$$

Writing $t = x_1 - v$, the integrand of $I(0)$ is proportional to

$$\frac{h(x_1)^2}{g(x_1)} p(x_1 | x_0) \propto \exp\left(t - e^{-t} + \frac{(t + \mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v))^2}{2\eta^2}\right).$$

We analyse the tails:

- *Right tail* $t \rightarrow +\infty$. The term $e^{-t} \rightarrow 0$, and expanding the quadratics the dominant behaviour is

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) + O(t)\right).$$

This is integrable if and only if the coefficient of t^2 is strictly negative, i.e.

$$\frac{1}{2\sigma^2} - \frac{1}{2\eta^2} < 0 \iff \eta^2 < \sigma^2.$$

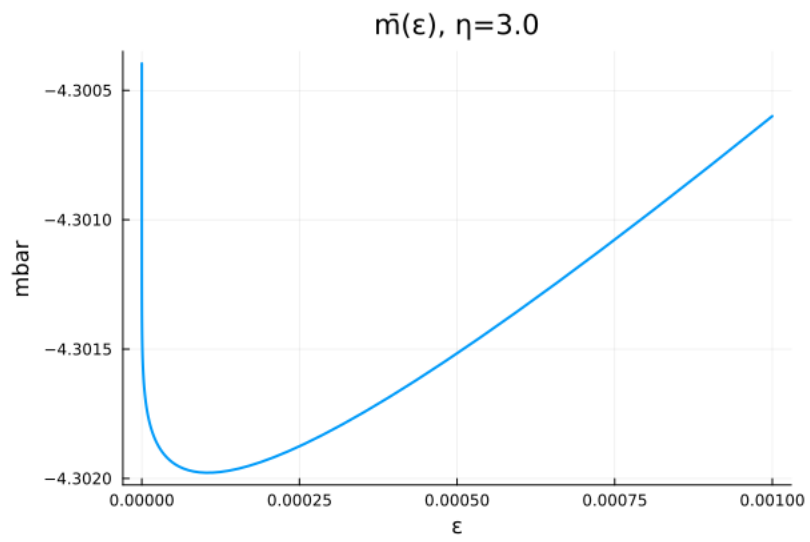
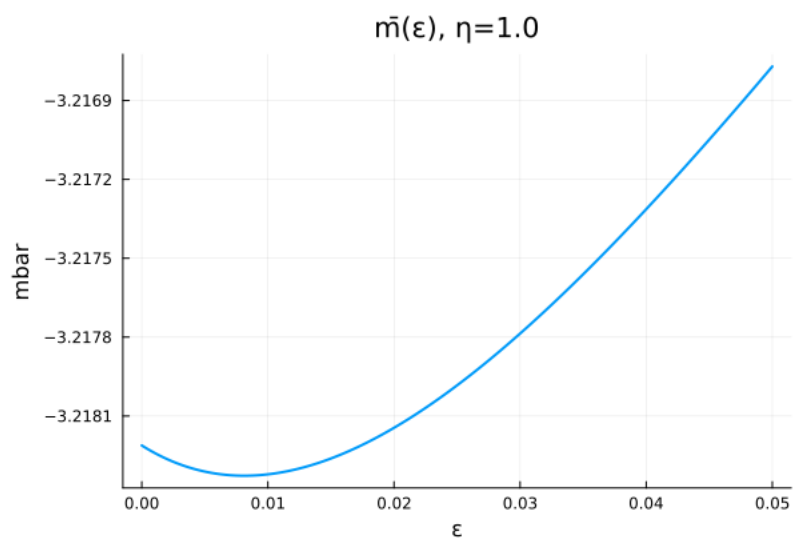
- *Left tail* $t \rightarrow -\infty$. The term $e^{-t} \rightarrow +\infty$ dominates inside the exponent, sending the integrand to zero faster than any polynomial. Hence the left tail never causes divergence.

Therefore,

Proposition 1. With the observation scheme $v = X_1 + \log Z^2$, $Z \sim \mathcal{N}(0, 1)$, we have

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1) < \infty \iff \eta < \sigma.$$

It follows that for $\eta < \pi/\sqrt{2} \approx 2.221$ no regularisation is needed. For $\eta = \sigma$ the integral diverges (though like e^t), while for $\eta > \sigma$ divergence is much faster.



3. IMPORTANCE SAMPLING QUALITY: KL AND χ^2 DIVERGENCES

3.1. General theory. The natural divergences for assessing IS quality of $\hat{L}$ are the KL divergence and the χ^2 divergence between the target P^h and the proposal $P^{g+\varepsilon}$:

$$\begin{aligned}\text{KL}(P^h \| P^{g+\varepsilon}) &= \int \log \frac{dP^h}{dP^{g+\varepsilon}} dP^h, \\ \chi^2(P^h \| P^{g+\varepsilon}) &= \int \left(\frac{dP^h}{dP^{g+\varepsilon}} \right)^2 dP^{g+\varepsilon} - 1.\end{aligned}$$

Since

$$\frac{dP^h}{dP^{g+\varepsilon}}(x_1) = \frac{h(x_1)/c_h}{(g(x_1) + \varepsilon)/(c_g + \varepsilon)} = \frac{G_\varepsilon(x_1)}{c_h},$$

we have

$$\chi^2(P^h \| P^{g+\varepsilon}) = \frac{m(\varepsilon)}{c_h^2} - 1,$$

so finite variance of the IS estimator is *exactly equivalent* to $\chi^2(P^h \| P^{g+\varepsilon}) < \infty$. The χ^2 divergence thus directly controls the variance, while the KL divergence controls the asymptotic quality of log-likelihood estimators (see Diaconis et al.).

3.2. KL divergence: explicit form. We have

$$\text{KL}(P^h \| P^{g+\varepsilon}) = \log \frac{c_g + \varepsilon}{c_h} + \int \log h(x_1) P^h(x_0, dx_1) - \int \log(g(x_1) + \varepsilon) P^h(x_0, dx_1).$$

As $\varepsilon \downarrow 0$ each term has a finite limit provided $\int |\log g(x_1)| P^h(x_0, dx_1) < \infty$. Since $\log g(x_1)$ is quadratic in x_1 (as g is Gaussian), this requires P^h to have finite second moments. Hence

$$\text{KL}(P^h \| P^{g+\varepsilon}) \rightarrow \text{KL}(P^h \| P^g) < \infty \quad \text{as } \varepsilon \downarrow 0,$$

whenever P^h has finite second moments, which holds for any $\varepsilon \geq 0$ in our specific example.

3.3. χ^2 divergence and finite variance: the threshold. For any $\varepsilon > 0$, the denominator $g(x_1) + \varepsilon \geq \varepsilon > 0$ uniformly, so

$$m(\varepsilon) \leq \varepsilon^{-1} \int h(x_1)^2 P(x_0, dx_1) < \infty,$$

provided $h \in L^2(P(x_0, \cdot))$, which holds for our specific h . Hence the IS estimator has finite variance for *every* $\varepsilon > 0$.

At $\varepsilon = 0$ the situation is more delicate. From the tail analysis of Section 2, the integrand $h(x_1)^2/g(x_1)$ behaves like

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) + O(t)\right) \quad \text{as } t = x_1 - v \rightarrow +\infty,$$

leading to the following result.

Proposition 2. For the observation scheme $v = X_1 + \log Z^2$, $Z \sim \mathcal{N}(0, 1)$:

$$\chi^2(P^h \| P^g) < \infty \iff m(0) < \infty \iff \eta < \sigma.$$

Now

$$ESS \approx \frac{N}{1 + \chi^2(P^h \| P^{g+\varepsilon})}.$$

So minimising $m(\varepsilon)$ at each SMC step is equivalent to maximising ESS.

4. NUMERICAL EXPERIMENT: EFFECT OF ε ON SMC PERFORMANCE

4.1. **Setup.** We consider the hidden Markov model

$$\begin{aligned} X_t \mid X_{t-1} &\sim \mathcal{N}(\omega + \psi X_{t-1}, \eta^2), \\ V_t \mid X_t &= X_t + \log Z_t^2, \quad Z_t \sim \mathcal{N}(0, 1), \end{aligned}$$

with parameters $\omega = 0$, $\psi = 0.9$, $\mu = -1.27$, $\sigma^2 = \pi^2/2$. We compare two regimes:

- **Regime A** ($\eta = 1.0 < \sigma \approx 2.22$): finite-variance regime, where $m(0) < \infty$.
- **Regime B** ($\eta = 3.0 > \sigma$): infinite-variance regime, where $m(0) = \infty$.

In both regimes we run SMC with the twisted proposal $P^{g+\varepsilon}$ for several values of $\varepsilon \in \{0, 0.001, 0.01, 0.05, 0.1\}$.

4.2. **SMC Algorithm.** At each time step t , the twisted proposal is the mixture

$$P^{g+\varepsilon}(x, \cdot) = \frac{c(x)}{c(x) + \varepsilon} P^g(x, \cdot) + \frac{\varepsilon}{c(x) + \varepsilon} P(x, \cdot),$$

where both $c(x)$ and the parameters of P^g are available in closed form (see Section 2). The IS weight for a particle x_t proposed from $P^{g+\varepsilon}(x_{t-1}, \cdot)$ is

$$G_\varepsilon(x_{t-1}, x_t) = \frac{h_t(x_t)}{g_t(x_t) + \varepsilon} (c(x_{t-1}) + \varepsilon).$$

To assess performance, we run Algorithm 1 independently $R = 200$ times on the same simulated dataset ($T = 50$, $N = 2000$) and record the minimum ESS over time,

$$\text{ESS}_{\min} = \min_{t=1, \dots, T} \text{ESS}_t,$$

for each run. A low $\text{ESS}_{\min}$ indicates that at some point the particle cloud collapsed to a near-degenerate distribution, with a single particle carrying almost all the weight.

4.3. **Results.** Figure 1 shows the empirical CDF of $\text{ESS}_{\min}$ across $R = 200$ runs for each value of ε , separately for the two regimes.

Regime A ($\eta < \sigma$). All curves are tightly clustered regardless of ε , with $\text{ESS}_{\min}$ concentrated between 400 and 600. This confirms the theoretical prediction: when $m(0) < \infty$, the IS weights have finite variance and ε has no meaningful effect on performance. Note though, while finiteness is guaranteed, it is typically beneficial to take nonzero ε . The value $m(0)$ can still be large, while finite.

Regime B ($\eta > \sigma$). The picture changes dramatically. At $\varepsilon = 0$ the CDF has substantial mass near zero, meaning that in a significant fraction of runs the particle cloud collapses to near-degeneracy at some time step. A small perturbation $\varepsilon = 0.001$ shifts the CDF markedly rightward, essentially eliminating the collapses. This is the direct numerical manifestation of $m(0) = \infty$: the IS weights have infinite variance, leading to occasional but catastrophic weight concentration on a single particle.

4.4. **On the difficulty of choosing ε in Regime B.** The results highlight that the choice of ε in Regime B is delicate. The optimal ε^* exists and can in principle be found by minimising $m(\varepsilon)$ numerically, as described in Section 3. However, $m(\varepsilon)$ depends on x_0 and varies across time steps as the particle cloud evolves, so a single fixed ε cannot be globally optimal. Furthermore, the performance is sensitive to this choice: the 5th

Algorithm 1 Guided particle filter (GPF) with twisted proposal $P^{g+\varepsilon}$, following ?, Algorithm 10.5

Require: observations $v_{1:T}$, particle count N , perturbation $\varepsilon \geq 0$, ESS threshold $\text{ESS}_{\min}$

- 1: For $n = 1, \dots, N$: draw $X_0^n \sim P(x_0)$, set $w_0^n \leftarrow G_0(X_0^n)$, $W_0^n \leftarrow w_0^n / \sum_{m=1}^N w_0^m$
 where $G_0(x_0) := P_0(dx_0) h_0(x_0) / M_0(x_0)$
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: **if** $\text{ESS}(W_{t-1}^{1:N}) < \text{ESS}_{\min}$ **then**
 - 4: $A_{t-1}^{1:N} \leftarrow \text{resample}(W_{t-1}^{1:N})$
 - 5: $\hat{w}_{t-1} \leftarrow 1$
 - 6: **else**
 - 7: $A_{t-1}^n \leftarrow n$
 - 8: $\hat{w}_{t-1} \leftarrow w_{t-1}^n$
 - 9: **end if**
 - 10: For $n = 1, \dots, N$: draw $X_t^n \sim M_t(X_{t-1}^{A_{t-1}^n}, dx_t) = P^{g_t+\varepsilon}(X_{t-1}^{A_{t-1}^n}, dx_t)$
 by sampling from $P^{g_t}(X_{t-1}^{A_{t-1}^n}, \cdot) = \mathcal{N}(\mu^*, \sigma^{*2})$ with probability $\alpha^n = c(X_{t-1}^{A_{t-1}^n}) / (c(X_{t-1}^{A_{t-1}^n}) + \varepsilon)$, and from $P(X_{t-1}^{A_{t-1}^n}, \cdot)$ otherwise
 - 11: $w_t^n \leftarrow \hat{w}_{t-1}^n \cdot G_t(X_{t-1}^{A_{t-1}^n}, X_t^n)$
 where

$$G_t(x_{t-1}, x_t) := \frac{h_t(x_t)}{g_t(x_t) + \varepsilon} \cdot (c(x_{t-1}) + \varepsilon)$$
 - 12: $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$
 - 13: **end for**
 - 14: **return** $\sum_{t=1}^T \log \ell_t^N$, $\{\text{ESS}(W_t^{1:N})\}_{t=1}^T$
 where, following equation (10.3) of ?,

$$\ell_t^N := \begin{cases} \frac{1}{N} \sum_{n=1}^N w_t^n & \text{if resampling occurred at time } t, \\ \left(\sum_{n=1}^N w_t^n \right) / \left(\sum_{n=1}^N w_{t-1}^n \right) & \text{otherwise.} \end{cases}$$
-

percentile of $\text{ESS}_{\min}$ drops sharply on both sides of ε^* , leaving a narrow window of well-performing values.

This suggests that a fixed ε is fundamentally limited in Regime B, and that an *adaptive* scheme — choosing ε_t at each step based on the current particle cloud by minimising an empirical estimate of $m(\varepsilon)$ — would be more robust. In Regime A, by contrast, $\varepsilon = 0$ is already optimal and no adaptation is needed.

5. ADAPTIVE CHOICE OF ε

5.1. Motivation. The numerical experiments of Section 4 show that the optimal perturbation ε^* depends on the current state of the particle cloud and varies across time steps. A fixed ε cannot be globally optimal, and performance is sensitive to this choice in the infinite-variance regime ($\eta > \sigma$). This motivates an adaptive scheme that selects ε_t^* at each step by minimising an empirical estimate of $m(\varepsilon)$ over the current particle cloud.

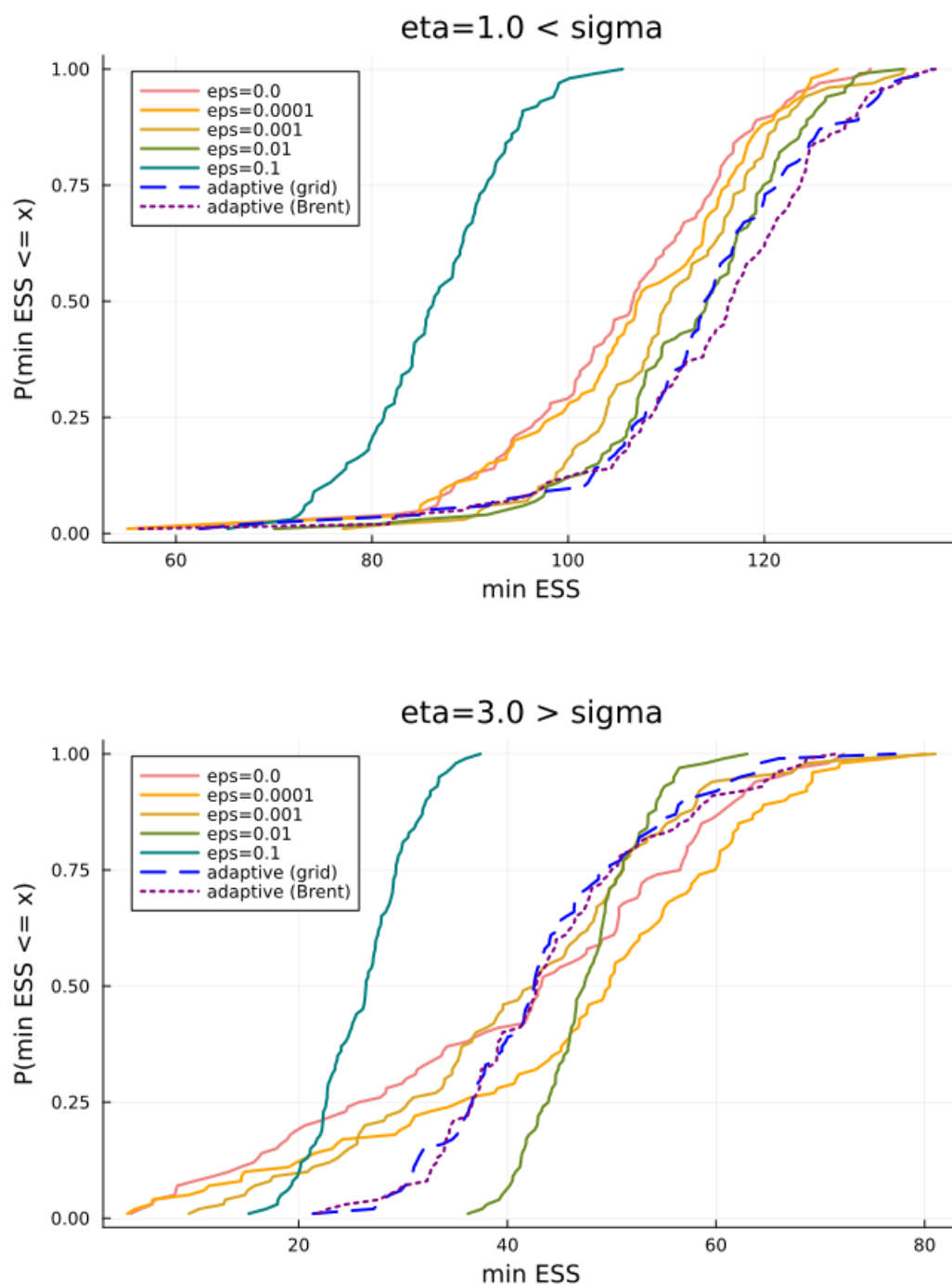


FIGURE 1.

5.2. **Empirical Estimate of $m(\varepsilon)$.** Recall that

$$m(\varepsilon) = (c(x_0) + \varepsilon) \int \frac{h(x_1)^2}{g(x_1) + \varepsilon} P(x_0, dx_1).$$

Given the particle cloud $\{x_{t-1}^{(i)}\}_{i=1}^N$ at time $t-1$ and observation v_t , we estimate $m(\varepsilon)$ by drawing auxiliary proposals $\{\tilde{x}_t^{(i)}\}_{i=1}^N$ from the *untwisted* kernel $P(x_{t-1}^{(i)}, \cdot)$ and computing

$$\hat{m}(\varepsilon) = (\bar{c}_t + \varepsilon) \cdot \frac{1}{N} \sum_{i=1}^N \frac{h_t(\tilde{x}_t^{(i)})^2}{g_t(\tilde{x}_t^{(i)}) + \varepsilon}, \quad \bar{c}_t = \frac{1}{N} \sum_{i=1}^N c(x_{t-1}^{(i)}).$$

Crucially, the auxiliary proposals $\tilde{x}_t^{(i)}$ are drawn independently of ε , so $\hat{m}(\varepsilon)$ can be evaluated over a grid of ε values at the cost of a single forward pass through P . The optimal perturbation is then

$$\varepsilon_t^* = \operatorname{argmin}_{\varepsilon \geq 0} \hat{m}(\varepsilon),$$

found by a search over a log-spaced grid $\varepsilon \in \{\varepsilon_1, \dots, \varepsilon_K\}$.

5.3. Adaptive SMC Algorithm.

Remarks.

- The auxiliary proposals $\{\tilde{X}_t^n\}$ in Step 2 are drawn from the untwisted kernel P and discarded after the grid search. Since they do not enter the weight update, the likelihood estimator $\sum_t \log \ell_t^N$ remains unbiased for $\log p(v_{1:T})$ for any data-dependent choice of ε_t^* .
- The grid $\mathcal{E}$ is log-spaced over $[10^{-6}, 1]$ with $K = 50$ points. All K evaluations of $\hat{m}(\varepsilon)$ reuse the same proposals $\{\tilde{X}_t^n\}$, so the cost of the grid search is one forward pass ($O(N)$) plus K evaluations of an $O(N)$ sum, totalling $O(KN)$ per step. As an alternative, one could minimise $\hat{m}(\varepsilon)$ using a gradient-based method, since the analytical derivative

$$\frac{d\hat{m}}{d\varepsilon} = \frac{1}{N} \sum_{n=1}^N \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \varepsilon} - (\bar{c}_t + \varepsilon) \frac{1}{N} \sum_{n=1}^N \frac{h_t(\tilde{X}_t^n)^2}{(g_t(\tilde{X}_t^n) + \varepsilon)^2}$$

is available in closed form, enabling Brent's method. However, in our Julia implementation with $N = 500$ and $T = 50$ steps, the grid search proved faster in practice:

Method	Total time (ms)	Time per step (ms)
Grid ($K = 50$)	23.26	0.465
Brent	40.81	0.816

The grid search is $1.75\times$ faster because the $K = 50$ evaluations form a tight vectorised loop, whereas Brent's method incurs per-iteration function-call overhead that outweighs the saving from fewer iterations (≈ 27 vs 50). Since $\hat{m}(\varepsilon)$ is smooth and unimodal, reducing to $K = 20$ or $K = 30$ grid points would roughly halve the cost with negligible impact on the quality of ε_t^* .

- In Regime A ($\eta < \sigma$), the minimiser ε_t^* is typically near the lower end of the grid, effectively recovering $\varepsilon_t^* \approx 0$ and the non-adaptive algorithm. In Regime B ($\eta > \sigma$), the minimiser is positive and varies over time, automatically adjusting to the current particle cloud.

Algorithm 2 Adaptive GPF with ε_t^* chosen by grid search

Require: observations $v_{1:T}$, particle count N , ESS threshold $\text{ESS}_{\min}$, grid $\mathcal{E} = \{\varepsilon_1 < \dots < \varepsilon_K\}$

- 1: For $n = 1, \dots, N$: draw $X_0^n \sim \mathcal{N}(0, \eta^2/(1 - \psi^2))$, set $W_0^n \leftarrow 1/N$, $w_0^n \leftarrow 1$
- 2: **for** $t = 1, \dots, T$ **do**
- 3: **Find** ε_t^* : draw auxiliary proposals $\tilde{X}_t^n \sim P(X_{t-1}^n, \cdot)$ for $n = 1, \dots, N$, and set

$$\varepsilon_t^* = \underset{\varepsilon \in \mathcal{E}}{\operatorname{argmin}} \hat{m}(\varepsilon), \quad \hat{m}(\varepsilon) = \left(\bar{c}_t + \varepsilon \right) \frac{1}{N} \sum_{n=1}^N \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \varepsilon}, \quad \bar{c}_t = \frac{1}{N} \sum_{n=1}^N c(X_{t-1}^n)$$

Discard $\{\tilde{X}_t^n\}$ after this step.

- 4: **if** $\text{ESS}(W_{t-1}^{1:N}) < \text{ESS}_{\min}$ **then**
- 5: $A_{t-1}^{1:N} \leftarrow \text{resample}(W_{t-1}^{1:N})$, $\hat{w}_{t-1}^n \leftarrow 1$
- 6: **else**
- 7: $A_{t-1}^n \leftarrow n$, $\hat{w}_{t-1}^n \leftarrow w_{t-1}^n$
- 8: **end if**
- 9: **Propose:** for $n = 1, \dots, N$ draw

$$X_t^n \sim P^{g_t + \varepsilon_t^*}(X_{t-1}^{A_{t-1}^n}, \cdot)$$

by sampling from $P^{g_t}(X_{t-1}^{A_{t-1}^n}, \cdot) = \mathcal{N}(\mu^*, \sigma^{*2})$ with probability $\alpha^n = c(X_{t-1}^{A_{t-1}^n}) / (c(X_{t-1}^{A_{t-1}^n}) + \varepsilon_t^*)$, and from $P(X_{t-1}^{A_{t-1}^n}, \cdot)$ otherwise

- 10: **Update weights:**

$$w_t^n = \hat{w}_{t-1}^n \cdot G_t(X_{t-1}^{A_{t-1}^n}, X_t^n), \quad G_t(x_{t-1}, x_t) := \frac{h_t(x_t)}{g_t(x_t) + \varepsilon_t^*} \cdot (c(x_{t-1}) + \varepsilon_t^*)$$

- 11: **Likelihood increment** (eq. 10.3 of ?):

$$\ell_t^N := \begin{cases} \frac{1}{N} \sum_{n=1}^N w_t^n & \text{if resampling occurred at time } t, \\ \frac{\sum_{n=1}^N w_t^n}{\sum_{n=1}^N w_{t-1}^n} & \text{otherwise} \end{cases}$$

- 12: **Normalise:** $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$
 - 13: **ESS:** $\text{ESS}_t \leftarrow (\sum_n (W_t^n)^2)^{-1}$
 - 14: **end for**
 - 15: **return** $\sum_{t=1}^T \log \ell_t^N$, $\{\text{ESS}_t\}_{t=1}^T$, $\{\varepsilon_t^*\}_{t=1}^T$
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5.4. Remarks.

Computational cost. The grid search adds one forward pass of N proposals from P and K evaluations of $\hat{m}(\varepsilon)$ per time step. Since h_t , g_t , and c are all available in closed form, the overhead is $O(NK)$ per step, which is negligible for moderate K (we use $K = 50$ log-spaced values).

Unbiasedness. The auxiliary proposals $\tilde{x}_t^{(i)}$ used to estimate ε_t^* are discarded after the grid search; the actual SMC proposals $x_t^{(i)}$ are drawn fresh from $P^{g_t+\varepsilon_t^*}$. This ensures that the IS estimator remains unbiased for $c_h(x_0)$ for any data-dependent choice of ε_t^* . Regime dependence. In Regime A ($\eta < \sigma$), the minimiser of $m(\varepsilon)$ is close to zero and the adaptive scheme essentially recovers $\varepsilon_t^* \approx 0$, giving performance comparable to the non-adaptive algorithm. In Regime B ($\eta > \sigma$), the adaptive scheme automatically selects a positive ε_t^* that prevents weight collapse, without requiring any prior knowledge of which regime the parameters fall in.

6. ADAPTIVE CHOICE OF ε VIA ESS TARGETING

6.1. Motivation. The numerical experiments of Section 4 show that the optimal perturbation ε^* varies across time steps and depends on the current particle cloud. A fixed ε cannot be globally optimal. We propose an adaptive scheme inspired by tempering methods in SMC ??, where a tempering parameter is chosen at each step so that the ESS after reweighting equals a target fraction ρN . The same principle applies here: we choose ε_t^* such that the ESS of the IS weights equals ρN .

6.2. ESS as a Function of ε . Given proposals $\{x_t^{(i)}\}_{i=1}^N$ drawn from $P(x_{t-1}^{(i)}, \cdot)$, the ESS of the weights $w^{(i)}(\varepsilon) = h_t(x_t^{(i)})/(g_t(x_t^{(i)}) + \varepsilon)$ is

$$\text{ESS}(\varepsilon) = \frac{(\sum_i w^{(i)}(\varepsilon))^2}{\sum_i w^{(i)}(\varepsilon)^2}.$$

Two key properties make this suitable for bisection:

- (1) $\text{ESS}(\varepsilon)$ is *increasing* in ε : as $\varepsilon \rightarrow \infty$, all weights converge to $h_t(x_t^{(i)})/\varepsilon$ and become equal, so $\text{ESS} \rightarrow N$.
- (2) $\text{ESS}(0)$ may be small or even zero if some $g_t(x_t^{(i)}) = 0$, reflecting the weight degeneracy in the infinite-variance regime.

The target ε_t^* is therefore the unique solution of

$$\text{ESS}(\varepsilon_t^*) = \rho N,$$

found by bisection on $\varepsilon \in [0, \varepsilon_{\max}]$. If $\text{ESS}(0) \geq \rho N$ already, we set $\varepsilon_t^* = 0$; if even $\varepsilon_{\max}$ is insufficient, we set $\varepsilon_t^* = \varepsilon_{\max}$.

6.3. Connection to $m(\varepsilon)$. The ESS targeting criterion is closely related to minimising $m(\varepsilon)$. Note that

$$\frac{\text{ESS}(\varepsilon)}{N} \approx \frac{1}{1 + \chi^2(P^h \| P^{g+\varepsilon})} = \frac{c_h(x_0)^2}{m(\varepsilon)},$$

so targeting $\text{ESS}(\varepsilon) = \rho N$ is equivalent to targeting $\chi^2(P^h \| P^{g+\varepsilon}) = (1 - \rho)/\rho$, i.e. a fixed level of the χ^2 divergence. Rather than minimising $m(\varepsilon)$ (which requires a grid search and gives a different target at each step), ESS targeting gives a directly interpretable, step-by-step criterion for particle diversity, and bisection finds the solution efficiently without a grid.

7. CONCLUSION

Figure ?? shows the results of a single SMC run for both regimes and both fixed ($\varepsilon = 0$) and adaptive ε^* . Even for $\eta = 3$ the results with $\varepsilon = 0$ can be just fine: a single run looks almost indistinguishable from the adaptive run in terms of ESS, cumulative log-likelihood, and incremental log-likelihood. However, if one were to repeat the analysis many times, occasionally this would result in a catastrophic drop in ESS. This is precisely the implication of $m(0) = \infty$: it does not mean every run fails, but that the variance of the IS weights across runs is infinite, so disasters occur with positive probability and cannot be ruled out from a single run alone. This is confirmed by the CDF plots of Figure 1, which required $R = 200$ independent runs to reveal the heavy left tail of the min-ESS distribution at $\varepsilon = 0$.

Two further observations support this interpretation. First, the incremental log-likelihood panel shows a pronounced dip around $t = 30$ in both the fixed and adaptive runs for $\eta = 3$. This dip is driven by a particular difficult observation and occurs regardless of ε , suggesting it reflects a genuinely hard step in the data rather than an algorithmic failure. In a catastrophic run, this dip would be far deeper and the ESS would collapse to near 1 at that point. Second, the adaptive ε^* panel for $\eta = 3$ shows ε^* spiking upward precisely around $t = 30$, at the same difficult observation. This confirms that the adaptive scheme is responding correctly: it detects the hard step through the empirical minimiser of $\hat{m}(\varepsilon)$ and automatically increases ε to protect against weight collapse, without any explicit knowledge of which observations are difficult.